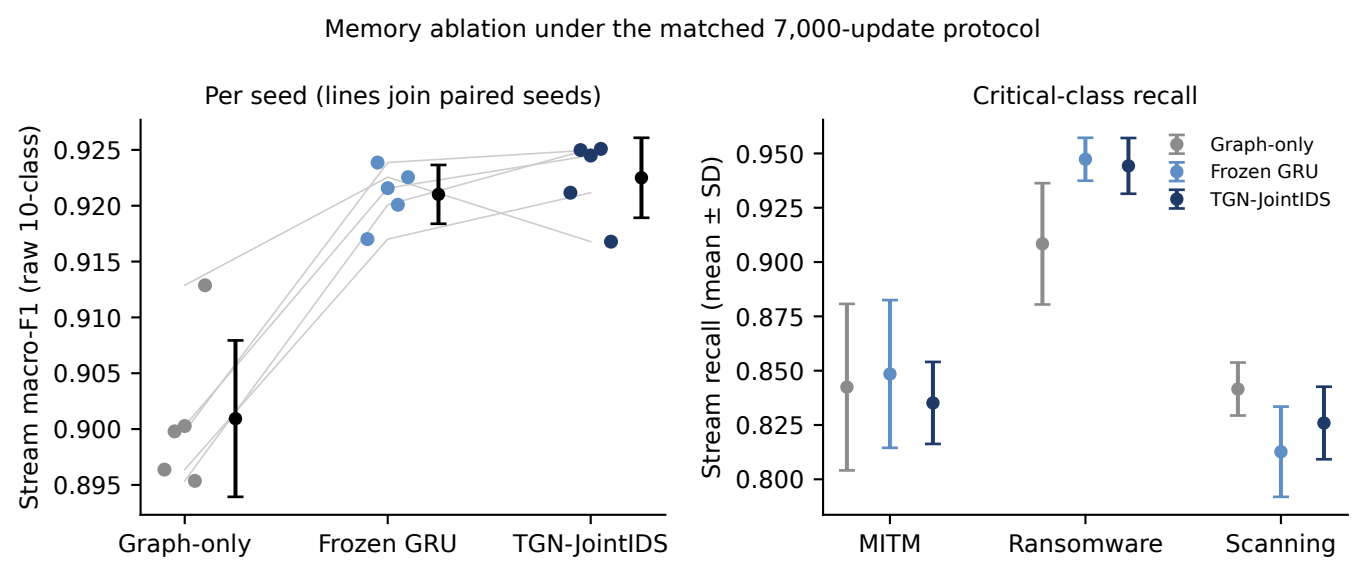
Memory ablation under the matched 7,000-update protocol
Per seed (lines join paired seeds)
Stream macro-F1 (raw 10-class)
0.925
0.920
0.915
0.910
0.905
0.900
0.895
Graph-only
Frozen GRU
TGN-JointIDS
Critical-class recall
Stream recall (mean ± SD)
0.950
0.925
0.900
0.875
0.850
0.825
0.800
Graph-only
Frozen GRU
TGN-JointIDS
MITM
Ransomware
Scanning